

FORM V    MATHEMATICS

Time allowed: 2 hours

Exam date: 14th May 2003

**Instructions:**

- All questions may be attempted.
- All questions are of equal value.
- All necessary working must be shown.
- Marks may not be awarded for careless or badly arranged work.
- Approved calculators and templates may be used.

**Collection:**

- Collect writing booklets in one bundle.
- Start each question in a new writing booklet.
- If you use a second booklet for a question, place it inside the first. Don't staple.
- Write your name, class, and master's initials on each writing booklet:

5P: JNC

5Q: REN

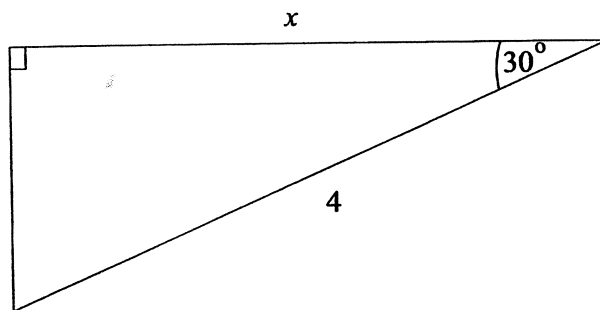
5R: BDD

**Checklist:**

- Folded A4 writing booklets required — 6 per boy.
- Candidature: 44 boys

QUESTION ONE (Start a new writing booklet)

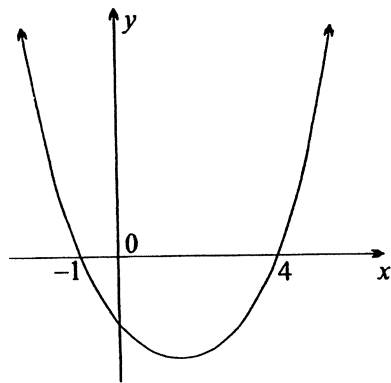
- (a) Solve  $5y - 1 = 14$ .
- (b) Expand and simplify  $(2 + \sqrt{5})^2$ .
- (c) Simplify  $\sqrt{98}$ .
- (d) Write down the exact value of  $\tan 120^\circ$ .
- (e) Write down the gradient of the line with equation  $3x + y + 7 = 0$ .
- (f) Express  $\frac{6}{\sqrt{3}}$  with a rational denominator.
- (g) Evaluate  $|2| - |-7|$ .
- (h) Solve the inequation  $2x - 1 \leq 5$  and graph the solution on a number line.
- (i) Simplify  $4\sqrt{3} + \sqrt{27}$ .
- (j) Factorise  $2x^2 + 7x - 15$ .
- (k) Find the value of  $f(2)$  if  $f(x) = x^2 + 3x - 1$ .
- (l)



In the diagram above, find the exact value of  $x$ .

**QUESTION TWO** (Start a new writing booklet)

- (a) Solve the equation  $|x - 4| = 3$ .
- (b) Express  $\frac{12}{\sqrt{5} - 2}$  with a rational denominator.
- (c) Sketch the graph of  $y = x + 1$  on a number plane.
- (d) Find rational numbers  $x$  and  $y$  such that  $x + 3\sqrt{y} = \sqrt{45} - 7$ .
- (e)

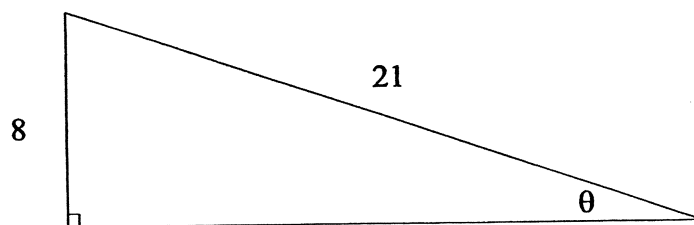


In the diagram above, the graph of  $y = x^2 - 3x - 4$  is drawn. Use the graph to solve the inequation  $x^2 - 3x - 4 \geq 0$ .

- (f) Solve the inequation  $|x + 2| < 4$ .
- (g) State the domain and range of the function  $f(x) = x^2$ .
- (h) Shade the region in the number plane in which  $y > 3 - x$ .

**QUESTION THREE** (Start a new writing booklet)

(a)



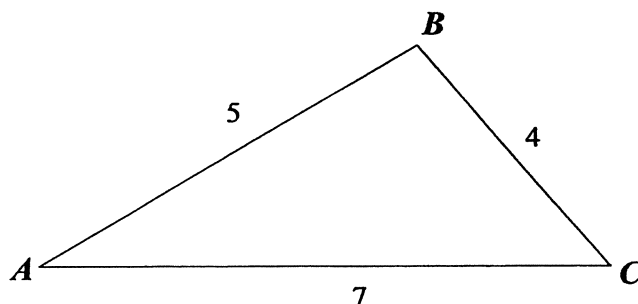
In the diagram above, find the value of  $\theta$ , correct to the nearest minute.

- (b) Solve  $\cos x = -\frac{\sqrt{3}}{2}$ , for  $0^\circ \leq x \leq 360^\circ$ .

(c) Simplify  $\frac{\tan \alpha}{\sec \alpha}$ .

(d) Given that  $\sin \theta = \frac{1}{3}$  and that  $\theta$  is obtuse, find the exact value of  $\tan \theta$ .

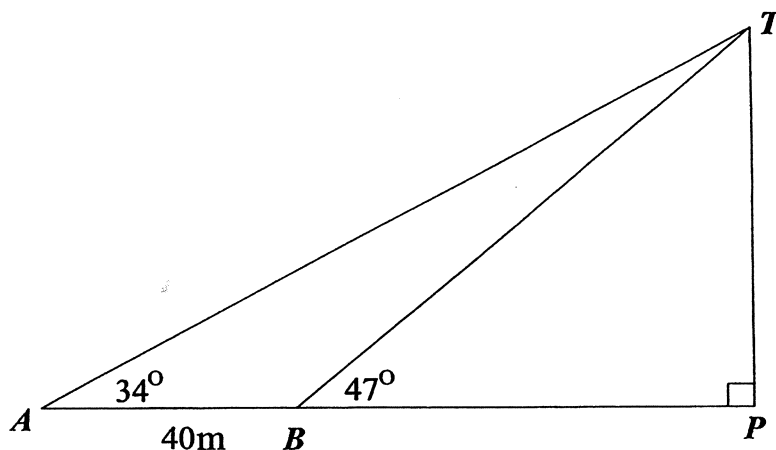
(e) (i)



In the diagram above, use the cosine rule to find the size of  $\angle ABC$ , correct to the nearest minute.

(ii) Find the area of  $\triangle ABC$ , correct to one decimal place.

(f)



In the diagram above, the angle of elevation of the top of the tower  $PT$  from  $A$  is  $34^\circ$ . From a point  $B$  40 metres closer to the tower, the angle of elevation is  $47^\circ$ .

(i) Show that  $\angle ATB = 13^\circ$ .

(ii) Use the sine rule to find the distance  $BT$ , correct to the nearest metre.

(iii) Hence find the height of the tower  $PT$ , correct to the nearest metre.

**QUESTION FOUR** (Start a new writing booklet)

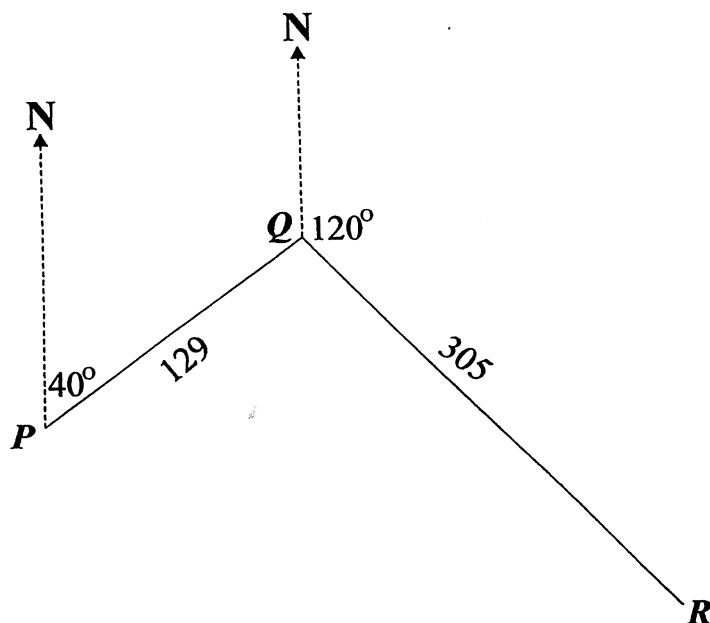
- (a) Write down the gradient of the line that makes an angle of  $30^\circ$  with the positive direction of the  $x$ -axis.
- (b) The point  $A$  has coordinates  $(-2, 1)$  and the point  $B$  has coordinates  $(-1, -1)$ .
  - (i) Find the distance  $AB$ .
  - (ii) Find the midpoint of the interval  $AB$ .
  - (iii) Find the gradient of the interval  $AB$ .
  - (iv) Find the equation of the line that passes through the points  $A$  and  $B$ .
- (c) Find, in general form, the equation of the line that has a gradient of  $-\frac{2}{3}$  and passes through the point  $(-2, 1)$ .
- (d) Find the equation of the line that passes through the origin and is perpendicular to the line  $2x - y + 9 = 0$ .
- (e) Show that the point  $(-1, 5)$  lies on the line  $3x + 2y - 7 = 0$ .
- (f) Find the perpendicular distance from the point  $(3, 5)$  to the line  $2x - y + 4 = 0$ .

**QUESTION FIVE** (Start a new writing booklet)

- (a) Sketch the following functions on SEPARATE number plane graphs showing intercepts, if any, and all other important features:
  - (i)  $3x - y + 6 = 0$ ,
  - (ii)  $y = (x - 3)(1 - x)$ , ( you do not need to give the coordinates of the vertex),
  - (iii)  $y = |x + 2|$ ,
  - (iv)  $y = -\frac{1}{x}$ .
- (b) Consider the function  $f(x) = \sqrt{1 - x^2}$ .
  - (i) Sketch the graph of  $y = f(x)$ .
  - (ii) State the domain of  $f(x)$ .
  - (iii) Explain why  $f(x)$  is an even function. (You may do this by referring to your graph or by giving an algebraic explanation.)
- (c) The equation of a circle is  $(x - 3)^2 + (y + 2)^2 = 16$ .
  - (i) Write down the coordinates of the centre of the circle and its radius.
  - (ii) Find the perpendicular distance from the line  $y = 2x + 3$  to the centre of the circle.
  - (iii) Explain why the line  $y = 2x + 3$  does not intersect the circle.

**QUESTION SIX** (Start a new writing booklet)

- (a) Solve  $\sin 2x = -\frac{1}{2}$ , for  $0^\circ \leq x \leq 180^\circ$ .
- (b) (i) Sketch the graphs of  $y = |3x - 1|$  and  $y = 1 - x$  on the same number plane.  
 (ii) Solve  $|3x - 1| = 1 - x$ .  
 (iii) Hence or otherwise, solve the inequation  $|3x - 1| > 1 - x$ .
- (c) Prove the identity  $\frac{1}{1 + \sin \theta} + \frac{1}{1 - \sin \theta} = 2 \sec^2 \theta$ .  
 (Start your proof:  $\text{LHS} = \frac{1}{1 + \sin \theta} + \frac{1}{1 - \sin \theta}$ )
- (d)



In the diagram above, a ship leaves Port  $P$  and sails 129 kilometres in a direction of  $040^\circ$  to Port  $Q$ . From Port  $Q$  the ship changes direction and sails 305 kilometres in a direction of  $120^\circ$  to Port  $R$ .

- (i) Explain why  $\angle PQR = 100^\circ$ .  
 (ii) How far is the ship then from Port  $P$ ? (Give your answer correct to the nearest kilometre.)  
 (iii) What is the bearing then of Port  $P$  from the ship? (Give your answer correct to the nearest degree.)

JNC

Question 1

(a)  $y = 3$

✓✓

(b)  $4 + 4\sqrt{5} + 5 = 9 + 4\sqrt{5}$

✓✓

(c)  $7\sqrt{2}$

✓

(d)  $-\sqrt{3}$

✓

(e)  $-3$

✓

(f)  $\frac{6\sqrt{3}}{3} = 2\sqrt{3}$

✓✓

(g)  $2 - 7 = -5$

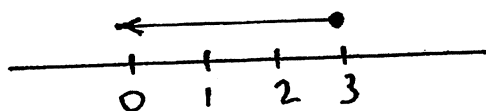
✓

(h)  $2x \leq 6$

✓✓

$x \leq 3$

✓



(i)  $4\sqrt{3} + 3\sqrt{3}$   
 $= 7\sqrt{3}$

✓

✓

(j)  $(2x - 3)(x + 5)$

✓

(k)  $f(2) = 2^2 + 3 \times 2 - 1$   
 $= 9$

✓

(l)  $\cos 30^\circ = \frac{x}{4}$

✓

$x = 4 \times \frac{\sqrt{3}}{2}$

$= 2\sqrt{3}$

✓

## Question 2

(a)  $x - 4 = 3$  or  $-(x - 4) = 3$

$\therefore x = 7$  or  $1$

✓✓

(1 each)

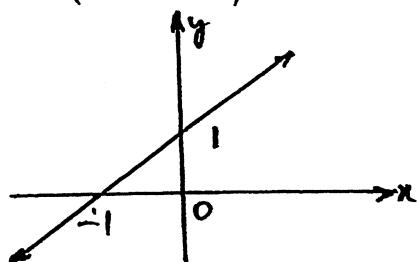
(b)  $\frac{12}{\sqrt{5} - 2} \times \frac{\sqrt{5} + 2}{\sqrt{5} + 2}$

✓

$= 12(\sqrt{5} + 2)$

✓

(c)



✓✓

(d)  $x = -7$ , and

✓

$3\sqrt{y} = \sqrt{45}$

✓

$3\sqrt{y} = 3\sqrt{5}$

$\therefore y = 5$

✓

(e)  $x \leq -1$  or  $x \geq 4$

✓✓

(1 each)

(f)  $x + 2 < 4$      $-(x + 2) < 4$

$x < 2$     ✓     $x + 2 > -4$

$x > -6$

✓✓

ie  $-6 < x < 2$

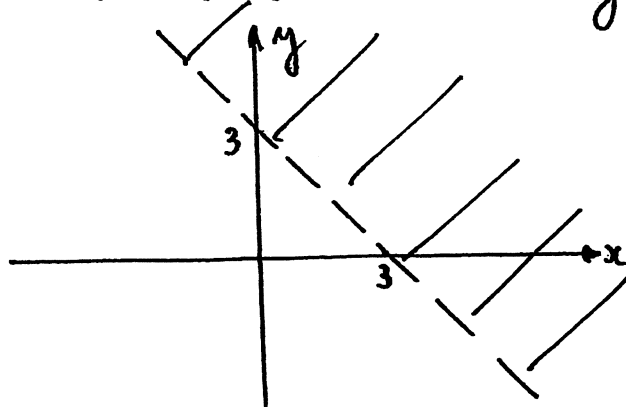
(not needed)

(g) Domain is all real  $x$  and range is  $y \geq 0$ .

✓✓

(1 each)

(h)



✓✓✓

1 line with intercepts

1 dotted

1 shading correct side

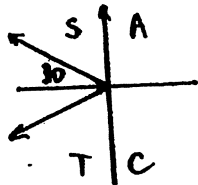
### Question 3

(a)  $\sin \theta = \frac{8}{21}$  ✓

$\theta = 22^\circ 24'$  ✓

(b)  $\cos^2 30^\circ + \sin^2 30^\circ$   
 $= \left(\frac{\sqrt{3}}{2}\right)^2 + \left(\frac{1}{2}\right)^2$   
 $= \frac{3}{4} + \frac{1}{4}$   
 $= 1$

(b) Related  $\angle = 30^\circ$  ✓

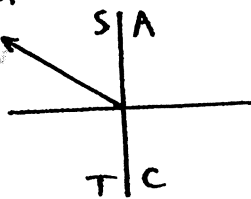
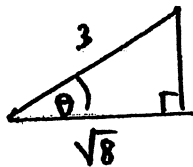


$x = 150^\circ \text{ or } 210^\circ$  ✓✓

(c)  $\frac{\frac{\sin \alpha}{\cos \alpha}}{\frac{1}{\cos \alpha}} = \frac{\sin \alpha}{\cos \alpha} \times \frac{\cos \alpha}{1}$

$= \sin \alpha$  ✓✓

(d)



$\tan \theta = -\frac{1}{\sqrt{8}}$  ✓✓

$\left( = -\frac{\sqrt{2}}{4} \right)$

(e)  $\cos B = \frac{4^2 + 5^2 - 7^2}{2 \times 4 \times 5}$  ✓  
 $= -\frac{1}{5}$  ✓

$\therefore \angle ABC = 101^\circ 32'$  ✓

(ii) Area  $\Delta = \frac{1}{2} \times 4 \times 5 \times \sin 101^\circ 32'$  ✓

$= 9.7979...$  ✓

$\approx 9.8$  ✓

(f)

(i)  $\angle ATB + 34 = 47$  (exterior  $\angle$  theorem) ✓  
 $\therefore \angle ATB = 13^\circ$

(ii)  $\frac{BT}{\sin 34} = \frac{40}{\sin 13}$  ✓

$BT = 99.4337...$  ✓

$\approx 99 \text{ m}$  ✓

(iii)  $\sin 47 = \frac{PT}{99}$

$\therefore PT = 99 \times \sin 47$  ✓

$= 72.404...$  ✓

$= 72 \text{ m}$  ✓

or 73.

1 for  $\sqrt{8}$

1 for -ve

(Wrong formula and correct from there and correct rounding 1/3)

(must be nearest minute for this mark)

(1/2 for wrong formula and no subsequent error)

(don't penalise rounding)

# Question 4

$$(a) m = \tan 30^\circ$$

$$= \frac{1}{\sqrt{3}}$$

✓

✓

$$(b) (i) \text{ distance} = \sqrt{1^2 + 2^2}$$

$$= \sqrt{5}$$

✓✓

$$(ii) \text{ Midpoint} = \left(-\frac{3}{2}, 0\right)$$

✓✓

$$(iii) m = -\frac{2}{1}$$

$$= -2$$

✓✓

$$(iv) y - 1 = -2(x + 2)$$

$$y = -2x - 3$$

✓

✓

(any form)

$$(c) y - 1 = -\frac{2}{3}(x + 2)$$

✓

$$3y - 3 = -2x - 4$$

✓

$$2x + 3y + 1 = 0$$

✓

(penalise if not in general form)

(d) gradient of line is 2. so

gradient of required line is  $-\frac{1}{2}$ . ✓

Eqn is:

$$y - 0 = -\frac{1}{2}(x - 0)$$

✓

$$2y = -x$$

$$x + 2y = 0$$

✓

$$(e) 3x + 2y - 7$$

$$= 3x - 1 + 2 \times 5 - 7$$

$$= -3 + 10 - 7$$

✓

$$= 0$$

No point lies on line.

$$(f) d = \frac{|6 - 5 + 4|}{\sqrt{2^2 + 1^2}}$$

✓

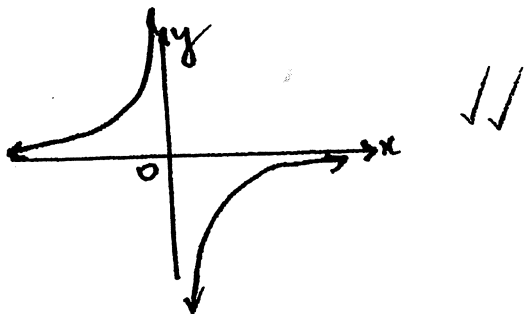
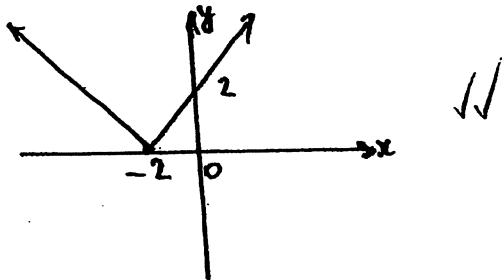
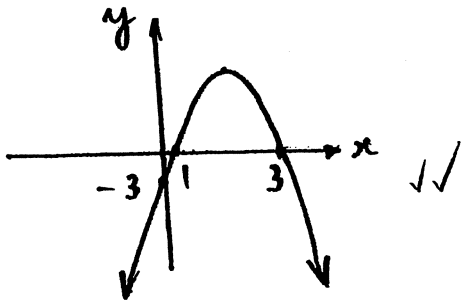
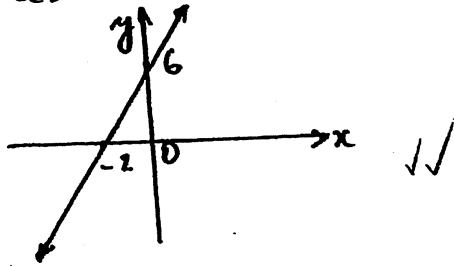
$$= \frac{5}{\sqrt{5}} = \sqrt{5}$$

✓

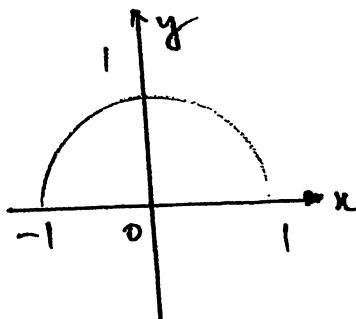
(no need to rationalise)

# Question 5

a)



(b)(i)



(ii) domain is  $-1 \leq x \leq 1$

(iii) symmetric in y axis, or

$$f(a) = \sqrt{1-a^2}$$

$$f(-a) = \sqrt{1-(-a)^2} = \sqrt{1-a^2} = f(a)$$

(c) (i) centre is  $(3, -2)$

$$r = 4$$

$$(ii) d = \frac{|3 \times 2 + (-2) \times (-1) + 3|}{\sqrt{1^2 + 2^2}}$$

$$= \frac{11}{\sqrt{5}}$$

$$= \frac{11\sqrt{5}}{5}$$

(iii)  $\frac{11\sqrt{5}}{5} \approx 4.9$  and so the

distance from the line to the centre of the circle is greater than the radius.

Graphs : -1 per error of omission of intercepts

0 marks if wrong shape

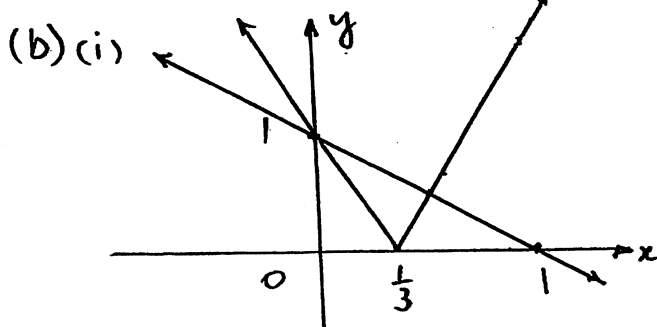
intercepts must be marked

### Question 6

(a) Related  $\angle = 30^\circ$  ✓

$2x = 210^\circ$  or  $330^\circ$  ✓

$x = 105^\circ$  or  $165^\circ$  ✓



✓ — line

✓✓ — absolute graph

(ii)  $|3x-1| = 1-x$

$3x-1 = 1-x$  or  $-(3x-1) = 1-x$

$4x = 2$

$2x = 0$

$x = \frac{1}{2}$  ✓✓

$x = 0$  ✓✓

(iii)  $x < 0$  or  $x > \frac{1}{2}$  ✓

(c) LHS =  $\frac{1}{1+\sin\theta} + \frac{1}{1-\sin\theta}$

$= \frac{1-\sin\theta + 1+\sin\theta}{1-\sin^2\theta}$  ✓

$= \frac{2}{\cos^2\theta}$

$= 2\sec^2\theta$  ✓

(d) (i)  $\angle PQR = 140^\circ$  (co-interior  $\angle$ ,  $\parallel$  lines)

$\angle PQR + 120 + 140 = 360$  ( $\angle$  in revolution) ✓✓

$\therefore \angle PQR = 100^\circ$

(ii)  $PR^2 = 129^2 + 305^2 - 2 \times 129 \times 305 \times \cos 100^\circ$  ✓

$= 123330.3751 \dots$

$\approx 351$  ✓

(iii)  $\frac{\sin R}{129} = \frac{\sin 100}{351}$

$\sin R = 0.3619 \dots$

$\therefore R \approx 21^\circ$  ✓

So bearing is  $270 + (90 - 60 - 21) = 279^\circ$  ✓